Almost sure spectral covariance escape for Gaussian ensemble Langevin discretization

Abstract

For every fixed positive step size and every nondegenerate Gaussian target in finite dimension, we prove that the largest empirical covariance eigenvalue of the untamed ensemble Langevin Euler scheme converges almost surely to positive infinity from every deterministic full-rank initial ensemble. The admissible ensemble size is $N \geq d+1$, and covariance is normalized by N. A measurable top eigenvector gives a Rayleigh lower bound with a conditionally standard Gaussian innovation. This bound is one-sided; the largest eigenvalue is not assigned a scalar Markov transition. A self-contained Gaussian logarithm lemma, including the two-particle scalar case, yields an inverse-moment contraction near zero. Conditional escape estimates and a martingale argument in the full particle filtration then establish the almost-sure limit. The source's rectangular covariance factor gives exact affine conjugacy. We separately prove rank preservation and spectral escape for nonzero singular ensembles under the same literal recurrence, outside the source's full-rank state space. No conclusion about lower eigenvalues, determinants, individual particles, centroids, or rates is asserted.

Keywords: ensemble Langevin discretization; Gaussian target; spectral covariance escape; almost sure divergence; rank preservation; stochastic particle methods

2020 Mathematics Subject Classification. Primary 60H35; Secondary 60F15, 39A50, 65C35.

1 The update and the spectral escape theorem

Fix finite integers $d \geq 1$ and $N \geq 2$, a step size $\eta > 0$, a vector $m \in \mathbb{R}^d$, and a symmetric positive-definite matrix $\Sigma \in \mathbb{R}^{d\times d}$. The target potential is

$$f(x) = \frac{1}{2}(x-m)^T\Sigma^{-1}(x-m). \tag{1}$$

For an ensemble $X = (x_1, \ldots, x_N)$ of column vectors in $\mathbb{R}^d$, define

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad Y = [x_1 - \bar{x} \ \cdots \ x_N - \bar{x}], \tag{2}$$

$$C = \frac{1}{N}YY^T, \qquad S = \frac{1}{\sqrt{N}}Y, \qquad SS^T = C. \tag{3}$$

All vector norms are Euclidean. The notation $\lambda_{\max}(C)$ denotes the largest eigenvalue of the symmetric positive-semidefinite matrix C.

Let all vectors $\xi_{n,i} \in \mathbb{R}^N$, $n \geq 0$ and $1 \leq i \leq N$, be independent standard Gaussians. We study exactly the untamed, fixed-step particle recurrence

$$\begin{aligned} x_{n+1,i} = x_{n,i} &- \eta C_n\Sigma^{-1}(x_{n,i} - m) \\ &+ \frac{\eta(d+1)}{N}(x_{n,i} - \bar{x}_n) + \sqrt{2\eta}\, S_n\xi_{n,i}. \end{aligned} \tag{4}$$

Here C_n and S_n are computed from the current ensemble by (3). In particular, S_n is the specified rectangular factor, not an unspecified symmetric square root.

Theorem 1 (Full-rank initial ensembles)**.** *Fix $d \geq 1$, $N \geq d+1$, $\eta > 0$, $m \in \mathbb{R}^d$, and $\Sigma \succ 0$. For every deterministic finite initial ensemble with $C_0 \succ 0$, the update* (4) *satisfies*

$$\mathbb{P}\big(x_{n,i} \text{ is finite for every finite } n, i,\ C_n \succ 0 \text{ for every finite } n\big) = 1, \tag{5}$$

$$\mathbb{P}\left(\lim_{n\to\infty} \lambda_{\max}(C_n) = +\infty\right) = 1. \tag{6}$$

Every assertion is for each fixed choice of parameters and initial ensemble. A simultaneous probability-one assertion over uncountably many choices is not intended.

The full-rank condition is the admissible state-space condition in Tan and Chen [1], pages 2 and 4. The recurrence (4) specializes their equations (13a)–(13b), on page 7, to (1). The covariance divisor N, finite-ensemble correction $(d+1)/N$, and N-dimensional particle noises are all retained. The source's stronger particle-count assumptions for continuous-time ergodicity are separate from the feasibility condition $N \geq d+1$ used here.

Theorem 2 (Nonzero singular ensembles under the literal recurrence)**.** *Fix $d \geq 1$, $N \geq 2$, $\eta > 0$, $m \in \mathbb{R}^d$, and $\Sigma \succ 0$. Define* (4) *algebraically from a deterministic finite ensemble whose covariance C_0 is nonzero, possibly singular. Then every finite-time ensemble is finite almost surely,*

$$\mathbb{P}\big(\operatorname{rank} C_n = \operatorname{rank} C_0 \text{ for every finite } n\big) = 1, \tag{7}$$

and (6) *holds. If $C_0 = 0$, then $C_n = 0$ for every n.*

Theorem 2 concerns the written recurrence beyond the full-rank state space. It does not place singular ensembles in that state space. A feasible covariance has rank at most $N-1$; no infeasible initial covariance is prescribed. The correction in (4) continues to use the ambient dimension d, even for a smaller covariance rank.

We prove the two theorems together. Sections 2–6 treat the standard target $m = 0$, $\Sigma = I_d$. Section 7 transfers the conclusions pathwise to every target (1). No claim is made for a degenerate target covariance, variable step sizes, non-Gaussian targets, tamed or regularized updates, or leave-one-out covariances.

2 Centering and the one sided spectral bound

For the standard target set

$$k = N-1, \qquad a = \frac{2\eta}{N}, \qquad b = \frac{d+1}{2} \geq 1, \qquad \gamma = \frac{\eta(d+1)}{N} = ba. \tag{8}$$

Thus d is the ambient particle dimension and k is the centered-particle coordinate dimension. They need not coincide. Write $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^N$ and choose a fixed $N \times k$ matrix Q such that

$$P = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T, \qquad Q^T Q = I_k, \qquad QQ^T = P. \tag{9}$$

Let Ξ_n be the $N \times N$ matrix with columns $\xi_{n,i}$. Its entries are independent standard Gaussians, independently over n. Define the full filtration

$$\mathcal{F}_n = \sigma(X_0, \Xi_0, \dots, \Xi_{n-1}), \qquad \mathcal{F}_\infty = \sigma\left(\bigcup_{n\geq 0} \mathcal{F}_n\right). \tag{10}$$

All current ensemble quantities are $\mathcal{F}_n$-measurable. Directly subtracting the new ensemble mean in (4) gives

$$Y_{n+1} = [(1+\gamma)I_d - \eta C_n]Y_n + \sqrt{2\eta}\, S_n \Xi_n P. \tag{11}$$

The projection P on the right of the noise is essential: the noise must also be centered.

2.1 Finite time rank preservation

All finite-time coordinates are finite by induction in the algebraic recurrence, since Gaussian entries are finite almost surely. To establish the rank assertion, condition on a current centered matrix Y of rank $r \geq 1$. Let U have orthonormal columns spanning range(Y). Necessarily $r \leq k$. Both the drift and noise in (11) lie in that range, because $C = YY^T/N$ and $S = Y/\sqrt{N}$.

The $r \times k$ matrix $U^T Y_+ Q$ has a Gaussian density on all of $\mathbb{R}^{r\times k}$. Indeed, ΞQ has independent standard Gaussian entries, and its multiplication by $\sqrt{2\eta}\, U^T S$ gives independent columns with covariance

$$2\eta\, U^T C U \succ 0.$$

Adding the deterministic drift does not change the existence of a density. Since $r \leq k$, the set of matrices of row rank less than r has Lebesgue measure zero: it is contained in the zero set of a fixed nonzero $r \times r$ minor polynomial. Thus $U^T Y_+ Q$ has row rank r almost surely. The identity $Y_+ = Y_+ P = Y_+ QQ^T$ and the range inclusion imply that Y_+ has rank exactly r almost surely.

This argument is conditional at each fixed state; it requires no globally measurable selection of the auxiliary basis U. Conditional induction and a countable union prove rank preservation at all finite times on one probability-one event. In particular, every C_n is nonzero whenever C_0 is nonzero, and full rank is retained from a full-rank start. If $C = 0$, then $Y = S = 0$, so the next centered update is zero as well.

2.2 A measurable top eigenvector and its Gaussian projection

A measurable unit top eigenvector can be chosen without assuming that the top eigenvalue is simple. For any nonzero positive-semidefinite C, let $r = \lambda_{\max}(C) > 0$ and put

$$\Pi(C) = \lim_{\ell \to \infty} (C/r)^{\ell}. \tag{12}$$

By finite-dimensional spectral calculus, this is the orthogonal projector onto the top eigenspace. Each entry is a pointwise limit of Borel functions of C. Choose the smallest index j for which $\|\Pi(C)e_j\| > 0$, where e_j are the standard basis vectors of $\mathbb{R}^d$, and set

$$v(C) = \frac{\Pi(C)e_j}{\|\Pi(C)e_j\|}.$$

The finite partition choosing j is Borel. Consequently v is a Borel selection, including at eigenvalue multiplicities; no continuity is asserted.

On the probability-one event established above define

$$R_n = \lambda_{\max}(C_n) > 0, \qquad v_n = v(C_n), \qquad u_n = \frac{Q^T Y_n^T v_n}{\sqrt{N R_n}}, \tag{13}$$

and

$$z_n = \frac{Q^T \Xi_n^T S_n^T v_n}{\sqrt{R_n}} \in \mathbb{R}^k. \tag{14}$$

The identity $Y_n P = Y_n$ gives $\|u_n\| = 1$. Conditionally on $\mathcal{F}_n$, the N coordinates of $\Xi_n^T S_n^T v_n / \sqrt{R_n}$ are independent standard Gaussians: the columns of Ξ_n are independent and

$$\|S_n^T v_n\|^2 = v_n^T C_n v_n = R_n.$$

Applying Q^T gives a standard Gaussian in $\mathbb{R}^k$. In particular this conditional law is constant, independently of the full past.

Taking the current top-vector Rayleigh quotient in (11) yields the exact identity

$$v_n^T C_{n+1} v_n = R_n \big\| (1 + \gamma - \eta R_n) u_n + \sqrt{a}\, z_n \big\|^2. \tag{15}$$

For example, this follows by multiplying the transpose of (11) by v_n, applying Q^T, and dividing its squared norm by N.

Fix $e = (1, 0, \ldots, 0)^T \in \mathbb{R}^k$. For a unit vector u, the Borel orthogonal matrix

$$O(u) = \begin{cases} I_k - 2\dfrac{(e-u)(e-u)^T}{\|e-u\|^2}, & u \neq e, \\ I_k, & u = e \end{cases}$$

satisfies $O(u)e = u$. Set $Z_n = O(u_n)^T z_n$. Its conditional law given $\mathcal{F}_n$ is standard Gaussian, so it is independent of $\mathcal{F}_n$. Moreover, Z_n is $\mathcal{F}_{n+1}$-measurable. Successive conditioning therefore proves that $(Z_n)_{n \geq 0}$ is iid standard Gaussian, rather than merely

marginally Gaussian. All these quantities may be defined arbitrarily on the common null exceptional set.

By norm invariance and the Rayleigh variational principle, (15) implies only

$$R_{n+1} \geq R_n W(R_n, Z_n), \qquad W(r,z) = \left\|(1+ba-\eta r)e + \sqrt{a}\, z\right\|^2. \tag{16}$$

This inequality, conditional on the full filtration, is the sole spectral comparison needed below. In particular, for $p > 0$ with the right side finite,

$$\mathbb{E}[R_{n+1}^{-p} \mid \mathcal{F}_n] \leq R_n^{-p}\mathbb{E}[W(R_n, Z)^{-p}], \qquad Z \sim \mathcal{N}(0, I_k). \tag{17}$$

The inverse power is decreasing, so the inequality reverses as required.

Remark. The exact formula (15) is not an exact recurrence for R_n. For instance, with $d = 2$, $N = 3$, $\eta = 1$, and $C_n = \operatorname{diag}(2,1)$, zero innovation gives $C_{n+1} = \operatorname{diag}(0,1)$. The old top-vector Rayleigh quotient is zero, whereas $\lambda_{\max}(C_{n+1}) = 1$. Such a centered ensemble exists; continuity gives the same strict distinction in a neighborhood of the zero innovation, which has positive Gaussian probability. No scalar Markov property of (R_n) will be used.

3 The Gaussian boundary logarithm

The small-state estimate rests on a positive logarithmic mean even when $k = 1$. We supply the analytic proof.

Lemma 1. *For a standard real Gaussian Z and every $a > 0$,*

$$G(a) := \mathbb{E}\log|1 + a + \sqrt{a}\, Z| > 0. \tag{18}$$

For every $b \geq 1$, $k \geq 1$, and standard Gaussian $Z \in \mathbb{R}^k$,

$$L := \mathbb{E}\log\left\|(1+ba)e + \sqrt{a}\, Z\right\|^2 > 0. \tag{19}$$

These logarithmic expectations are finite.

Proof. Let ϕ be the standard real Gaussian density and put

$$J(x) = \mathbb{E}\log|x + Z| = \int_{\mathbb{R}} \log|t|\phi(t-x)\, dt, \qquad H = J'.$$

The logarithmic singularity is locally integrable, and the tails are controlled by Gaussian decay. On each compact set of shifts x, derivatives of the density times $|\log|t||$ admit common integrable bounds. Differentiating the density under the integral therefore shows that J is smooth. Symmetry gives $H(0) = 0$.

Direct differentiation of the density gives

$$J''(x) + xJ'(x) = \int_{\mathbb{R}} \log|t|[t(t-x)-1]\phi(t-x)\,dt = 1. \tag{20}$$

To justify the last equality, the integrand is $-\log|t|\,(d/dt)[t\phi(t-x)]$. Integrate by parts outside $(-\varepsilon,\varepsilon)$. The result is $\int_{|t|>\varepsilon}\phi(t-x)\,dt$ plus boundary terms of order $\varepsilon|\log\varepsilon|$, which vanish. Hence $H' + xH = 1$ and

$$H(x) = e^{-x^2/2}\int_0^x e^{t^2/2}\,dt. \tag{21}$$

In particular, $H(x) > 0$ when $x > 0$, so J is strictly increasing there.

We next prove

$$H(x) < \frac{x}{x^2-2} < \frac{1}{\sqrt{x^2-4}}, \qquad x > 2. \tag{22}$$

For $D(x) = x/(x^2-2) - H(x)$, direct differentiation yields

$$D'(x) + xD(x) = \frac{x^2-6}{(x^2-2)^2}. \tag{23}$$

At $x = \sqrt{6}$, the inequality $D(x) > 0$ is, by (21), equivalent to

$$\frac{e^3}{4} - \int_0^1 e^{3u^2}\,du > 0. \tag{24}$$

Termwise integration of the uniformly convergent exponential series expresses the left side as

$$\sum_{j=0}^{\infty}\frac{3^j}{j!}\left(\frac{1}{4}-\frac{1}{2j+1}\right).$$

The first five terms have the exact positive sum

$$-\frac{3}{4}-\frac{1}{4}+\frac{9}{40}+\frac{27}{56}+\frac{15}{32} = \frac{197}{1120} > 0, \tag{25}$$

and every later term is positive. Thus (24) holds. By (23), the derivative of $e^{x^2/2}D(x)$ is negative on $(2,\sqrt{6})$ and positive on $(\sqrt{6},\infty)$. Integrating backward or forward from its positive value at $\sqrt{6}$ proves $D(x) > 0$ for all $x > 2$. The second inequality in (22) follows by squaring positive quantities and using

$$(x^2-2)^2 - x^2(x^2-4) = 4.$$

Set $x(a) = \sqrt{a} + 1/\sqrt{a}$. Then

$$\begin{aligned} G(a) &= \frac{1}{2}\log a + J(x(a)), \\ G'(a) &= \frac{1}{2a} + \frac{a-1}{2a^{3/2}}H(x(a)). \end{aligned} \tag{26}$$

For $a \geq 1$ the derivative is positive, with value $1/2$ at $a = 1$. For $0 < a < 1$, one has $x(a) > 2$ and

$$\frac{1}{\sqrt{x(a)^2 - 4}} = \frac{\sqrt{a}}{1 - a}.$$

Use (22) in (26). The coefficient $a - 1$ is negative, so multiplying reverses the strict inequality and again gives $G'(a) > 0$.

It remains to verify $G(a) \to 0$ as $a \downarrow 0$ despite the logarithmic pole. Write $V_a = 1 + a + \sqrt{a}\, Z$. On $|V_a| \geq 1/2$, the negative logarithm is bounded; the positive logarithm is uniformly integrable for $0 < a \leq 1$, since $\log^+ |V_a| \leq |V_a|$ and $\mathbb{E}|V_a|^2$ is bounded. As $V_a \to 1$ in probability, this part of the expectation tends to zero. On $|y| < 1/2$, the density of V_a is bounded by

$$(2\pi a)^{-1/2} \exp\left(-\frac{1}{8a}\right).$$

Its product with the finite integral $\int_{-1/2}^{1/2} |\log |y||\, dy$ tends to zero. Therefore $G(a) \to 0$, and strict monotonicity proves (18).

For $b \geq 1$, monotonicity of J on positive arguments now gives

$$\begin{aligned} \mathbb{E} \log |1 + ba + \sqrt{a}\, Z| &= \frac{1}{2} \log a + J\left(\frac{1 + ba}{\sqrt{a}}\right) \\ &\geq G(a) > 0. \end{aligned} \tag{27}$$

In k dimensions, the norm-squared logarithm is integrable near zero by bounded density and radial logarithmic integrability, and at infinity by Gaussian tails. Taking e to be the first coordinate vector, the pointwise norm bound and (27) imply

$$L \geq 2\mathbb{E} \log |1 + ba + \sqrt{a}\, Z_1| > 0.$$

This proves (19) for every $k \geq 1$, including $N = 2$. ∎

4 Excluding eventual trapping near zero

For the function W in (16), write

$$M(p, r) = \mathbb{E}[W(r, Z)^{-p}], \qquad Z \sim \mathcal{N}(0, I_k).$$

Choose p_0 with $0 < p_0 < k/2$. A bounded Gaussian density near the origin makes

$$\|y\|^{-2p_0} \big(1 + |\log \|y\|^2|\big)$$

locally integrable there: in radial coordinates the power is $k - 1 - 2p_0 > -1$. On the complement of the unit ball, Gaussian tails control the logarithmic factors. These bounds justify differentiation from the right at $p = 0$, yielding

$$M(0, 0) = 1, \qquad \partial_p M(0, 0) = -\mathbb{E} \log W(0, Z) = -L < 0. \tag{28}$$

Consequently some fixed $p \in (0, k/2)$ satisfies $M(p, 0) < 1$.

Continuity of $M(p, r)$ in r follows by writing its integral with the singular factor $\|y\|^{-2p}$ fixed and moving the Gaussian density. Shifts in a bounded neighborhood have a common bounded-density estimate near zero and a common Gaussian envelope at infinity. Dominated convergence applies. Hence there are constants $\delta > 0$ and $q \in (0, 1)$ such that

$$M(p, r) \le q \qquad (0 \le r \le \delta). \tag{29}$$

These constants are uniform over all compatible ensemble configurations. From (17),

$$\mathbb{E}[R_{n+1}^{-p} \mid \mathcal{F}_n] \le q R_n^{-p} \quad \text{on } \{0 < R_n \le \delta\}. \tag{30}$$

Fix a deterministic time $m \ge 0$ and let

$$T_m = \inf\{n \ge m : R_n > \delta\}.$$

For $j \ge 0$, dropping only nonnegative killed terms and iterating (30) gives, on $\{R_m \le \delta\}$,

$$\mathbb{E}\big[R_{m+j}^{-p} \mathbf{1}_{\{T_m > m+j\}} \mid \mathcal{F}_m\big] \le q^j R_m^{-p}. \tag{31}$$

Indeed, in the step from j to $j+1$, the indicator of $T_m > m + j + 1$ is at most that of $T_m > m + j$; on the latter event $R_{m+j} \le \delta$, so conditional expectation at time $m + j$ supplies the factor q.

On the survival event in (31), $R_{m+j}^{-p} \ge \delta^{-p}$. Therefore

$$\mathbb{P}(T_m > m + j \mid \mathcal{F}_m) \le \delta^p q^j R_m^{-p} \longrightarrow 0 \quad \text{on } \{R_m \le \delta\}. \tag{32}$$

The limit holds almost surely because $R_m > 0$ and is finite. Conditional expectations of nonnegative variables may initially be extended-valued; no assumption that $\mathbb{E}R_m^{-p} < \infty$ is needed. Conditional monotone convergence applied to the decreasing survival events gives $\mathbb{P}(T_m = \infty \mid \mathcal{F}_m) = 0$. On $\{R_m > \delta\}$, $T_m = m$ already.

For each deterministic m, the event that every R_n with $n \ge m$ is at most δ thus has probability zero. A countable union over m proves

$$\mathbb{P}(R_n > \delta \text{ infinitely often}) = 1. \tag{33}$$

The bound (32) depends on the present value through R_m^{-p}; a uniform finite exit-time estimate as $R_m \downarrow 0$ is not being claimed.

5 Uniform conditional escape probabilities

Define the event

$$A = \{R_n \longrightarrow +\infty\}. \tag{34}$$

Choose a finite $B > \delta$ large enough that

$$B \geq \frac{2(1+ba)}{\eta}, \qquad \frac{\eta^2 B^2}{16} \geq 2, \qquad \frac{64ak}{3\eta^2 B^2} \leq \frac{1}{2}. \tag{35}$$

Such a choice exists for every fixed $\eta > 0$, d, and N.

Fix a deterministic time n. On $\{R_n \geq B\}$, consider the events

$$E_j = \left\{ \|Z_{n+j}\| \leq \frac{\eta 2^j B}{4\sqrt{a}} \right\}, \qquad j \geq 0. \tag{36}$$

If $E_0, \ldots, E_{j-1}$ hold, we claim inductively that $R_{n+j} \geq 2^j B$. For such a state, the first inequality in (35) implies

$$|1 + ba - \eta R_{n+j}| \geq \frac{\eta}{2} R_{n+j}.$$

On E_j, the noise norm inside W is at most $\eta 2^j B/4 \leq \eta R_{n+j}/4$. The reverse triangle inequality in (16) consequently gives

$$R_{n+j+1} \geq \frac{\eta^2}{16} R_{n+j}^3 \geq 2R_{n+j} \geq 2^{j+1} B. \tag{37}$$

This proves the induction. Thus $\bigcap_{j\geq 0} E_j$ implies actual convergence of the original spectral process to infinity whenever $R_n \geq B$.

For each $j \geq 0$, the conditional law of Z_{n+j} given $\mathcal{F}_n$ is standard Gaussian, by the tower property applied first at time $n+j$. Markov's inequality and a countable conditional union bound yield

$$\begin{aligned} \mathbb{P}\left(\bigcup_{j\geq 0} E_j^c \,\middle|\, \mathcal{F}_n \right) &\leq \sum_{j\geq 0} \frac{16ak}{\eta^2 4^j B^2} \\ &= \frac{64ak}{3\eta^2 B^2} \leq \frac{1}{2}. \end{aligned} \tag{38}$$

Conditional independence is not needed for this estimate. We have shown, for every deterministic n, that

$$\mathbb{P}(A \mid \mathcal{F}_n) \geq \frac{1}{2} \quad \text{on } \{R_n \geq B\}. \tag{39}$$

The pathwise lower growth bound (37) is used only on the constructed event; it is not an almost-sure escape-rate assertion.

For the remaining interval, set

$$D = \max_{\delta \leq r \leq B} |1 + ba - \eta r|, \qquad K = \frac{D + \sqrt{B/\delta}}{\sqrt{a}}, \qquad \beta = \mathbb{P}(\|Z\| \geq K) > 0. \tag{40}$$

The threshold is finite and a Gaussian has positive tail probability at every finite threshold. If $\delta < R_n < B$ and $\|Z_n\| \geq K$, then

$$\left\| (1 + ba - \eta R_n)e + \sqrt{a}\, Z_n \right\| \geq \sqrt{a}\, \|Z_n\| - D \geq \sqrt{B/\delta}.$$

The lower bound (16) therefore implies $R_{n+1} \geq B$, and

$$\mathbb{P}(R_{n+1} \geq B \mid \mathcal{F}_n) \geq \beta \quad \text{on } \{\delta < R_n < B\}. \tag{41}$$

Apply (39) at time $n+1$ and condition again:

$$\begin{aligned} \mathbb{P}(A \mid \mathcal{F}_n) &= \mathbb{E}[\mathbb{P}(A \mid \mathcal{F}_{n+1}) \mid \mathcal{F}_n] \\ &\geq \frac{1}{2}\mathbb{P}(R_{n+1} \geq B \mid \mathcal{F}_n) \geq \frac{\beta}{2} \quad \text{on } \{\delta < R_n < B\}. \end{aligned} \tag{42}$$

Combining (39) and (42), since $\beta \leq 1$, gives the uniform positive bound

$$\mathbb{P}(A \mid \mathcal{F}_n) \geq \varepsilon := \frac{\beta}{2} > 0 \quad \text{on } \{R_n > \delta\}. \tag{43}$$

Every estimate is conditional on the entire ensemble history. No escape-probability function of a scalar Markov state is introduced.

6 The full filtration martingale

The event A is $\mathcal{F}_\infty$-measurable. For example,

$$A = \bigcap_{h=1}^{\infty} \bigcup_{m=0}^{\infty} \bigcap_{n \geq m} \{R_n > h\}.$$

Thus the bounded martingale $M_n = \mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_n]$ satisfies

$$M_n \longrightarrow \mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_\infty] = \mathbf{1}_A \quad \text{almost surely} \tag{44}$$

by upward convergence of conditional expectations. The relevant property is measurability in $\mathcal{F}_\infty$; no tail zero-one law is required.

Take the common probability-one event on which (33), (44), and the countably many bounds (43) all hold. Along each such path, $R_n > \delta$ infinitely often, and hence $M_n \geq \varepsilon$ infinitely often. Its limit cannot be zero. Since the limit is $\mathbf{1}_A$, it must be one. Therefore

$$\mathbb{P}(A) = 1.$$

This establishes convergence to positive infinity, not merely an unbounded limsup, for every nonzero initial covariance in the standard-target recurrence. Together with Section 2, it proves the standard-target versions of both theorems.

7 Affine conjugacy for Gaussian targets

Let $L = \Sigma^{-1/2}$, the fixed symmetric positive-definite inverse square root, and define transformed particles

$$y_{n,i} = L(x_{n,i} - m).$$

The transformed ensemble quantities satisfy

$$\bar{y}_n = L(\bar{x}_n - m), \qquad C_n^y = LC_n^x L^T, \qquad S_n^y = LS_n^x. \tag{45}$$

The last identity is exact because the covariance factor is the rectangular centered-particle matrix in (3). It does not rely on a generally false equivariance identity for a preferred symmetric covariance square root.

Multiplying (4) by L and subtracting Lm gives, with the same original Gaussian vectors $\xi_{n,i}$,

$$\begin{aligned} y_{n+1,i} = y_{n,i} &- \eta C_n^y y_{n,i} \\ &+ \frac{\eta(d+1)}{N}(y_{n,i} - \bar{y}_n) + \sqrt{2\eta}\, S_n^y \xi_{n,i}. \end{aligned} \tag{46}$$

Indeed,

$$LC_n^x \Sigma^{-1}(x_{n,i} - m) = LC_n^x L^T y_{n,i} = C_n^y y_{n,i}.$$

Thus (46) is pathwise exactly the standard-target recurrence, with the same η, d, N, rather than merely a distributional comparison. Finiteness and covariance rank are preserved under this fixed invertible affine change.

Finally, $C_n^x = \Sigma^{1/2} C_n^y \Sigma^{1/2}$ implies

$$\begin{aligned} \lambda_{\min}(\Sigma)\lambda_{\max}(C_n^y) &\le \lambda_{\max}(C_n^x) \\ &\le \lambda_{\max}(\Sigma)\lambda_{\max}(C_n^y). \end{aligned} \tag{47}$$

For a direct proof, write $\lambda_{\max}(C_n^x) = \|\Sigma^{1/2}(C_n^y)^{1/2}\|_{\mathrm{op}}^2$. The upper bound is submultiplicativity; the lower bound follows from $\|\Sigma^{1/2} w\| \ge \sqrt{\lambda_{\min}(\Sigma)}\|w\|$. These statements hold also when C_n^y is singular. Both target-dependent constants are fixed and strictly positive. The standard-target spectral escape therefore transfers to C_n^x. If $C_0^x = 0$, the centered particles and rectangular noise factor vanish and remain zero. This completes both proofs.

8 Source comparison and scope

The reference is specifically arXiv:2609.03326v1 [1]. Its page 1 uses divisor N in the empirical covariance and the correction $(d+1)/N$ in equation (3). Pages 2 and 4 describe the full-rank admissible state space. Page 7, equations (13a)–(13b), specifies the untamed Euler update and its rectangular noise factor. These conventions are exactly those used in (4).

Section 3.1 of the source, on pages 7–8, gives positive-probability covariance divergence for the one-dimensional standard Gaussian target from sufficiently large initial covariance, for each fixed positive step size. The present result strengthens that conclusion to almost-sure largest-eigenvalue escape from every deterministic full-rank start, in every finite dimension and for every nondegenerate Gaussian target. The singular extension is explicitly about the same literal recurrence on an enlarged domain.

Source equation (16) is an inequality on a specified small-noise event, not an unconditional scalar covariance transition. The present proof does not replace that inequality by equality. In multiple dimensions it uses only the current top-vector Rayleigh identity and the one-sided bound (16). The proof is self-contained apart from standard Gaussian facts, elementary finite-dimensional linear algebra, and conditional-expectation convergence; no numerical experiment or unproved scalar logarithm estimate is needed.

This result is consistent with the source's direct-Euler divergence analysis. It does not refute its continuous-time ergodicity results or its theorems for the covariance-trace regularized dynamics and discretizations. Those are different algorithms or time regimes with their own hypotheses. No literature-priority claim is made.

Spectral escape concerns only $\lambda_{\max}(C_n)$. The proof gives no divergence assertion for $\lambda_{\min}(C_n)$ or $\det C_n$, no specified-particle or centroid conclusion, and no almost-sure rate. Preservation of a positive finite-time rank does not supply such conclusions about the asymptotic behavior of other eigenvalues. The argument also makes no claim for adaptive or vanishing step sizes, degenerate Gaussian targets, non-Gaussian potentials, regularized or tamed updates, or leave-one-out constructions.

References

[1] H. Y. Tan and Y. Chen. Geometric ergodicity of affine invariant ensemble Langevin and its discrete time variants. *arXiv preprint arXiv:2609.03326v1*, 2026. `https://arxiv.org/abs/2609.03326v1`.